



GT/LowE/SP5 Energy Systems – Digital Solutions

Electro-mechanical interactions in offshore wind turbines

AUTHORS

August Norius Høidalen

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Abstract

This work investigates electromechanical interactions between an offshore wind turbine and a weak, islanded power system using the LEOGO network as a representative offshore grid. A reduced-order wind turbine model and a high-fidelity OpenFAST model are coupled to the electrical network to study how grid disturbances propagate into the turbine drivetrain and tower dynamics. Frequency-support control based on active-power droop and synthetic inertia is implemented, enabling the turbine to contribute to system frequency regulation. Simulations show that droop improves the frequency nadir following a generation loss, while synthetic inertia primarily reduces the initial rate of change of frequency. However, the same control action introduces an electromechanical coupling path through generator torque, allowing grid disturbances near the tower side-to-side natural frequency to excite significant structural oscillations. A notch filter tuned to this mode is shown to suppress the resulting tower excitation while retaining the frequency-support response. The results highlight the importance of considering mechanical turbine dynamics when designing frequency-support controllers for converter-interfaced offshore wind turbines.

Introduction

1 The LEOGO network

The Low Emission Oil and Gas Open (LEOGO) platform is a *hypothetical but realistic offshore oil & gas installation defined by SINTEF (Svendsen et al., IET Energy Systems Integration, 2023)* KILDE as an **open reference case** for studyingh renewable integration into offshore ("off-grid") energy systems. All data is public and non-confidential, which is why we use it, so our results can be reproduced and compared. It is essentially an **isolated micro-grid** that is always in island mode, with a peak electrical demand of about 47 MW.

What makes it distinctive

- **33 kV** - wind turbine collection grid and transmission to the platform.
- **11 kV** - main busbar; gas turbines and large loads connect here. Realised as a split single busbar, normally closed tie breaker.
- **690 V** - utility busbar (smaller pumps/aux equipment), fed from 11 kV.
- **400 V** - accommodation loads, fed from 690 V.
- **+1300 V DC** - drilling switchboard, fed via rectifiers from 11 kV.

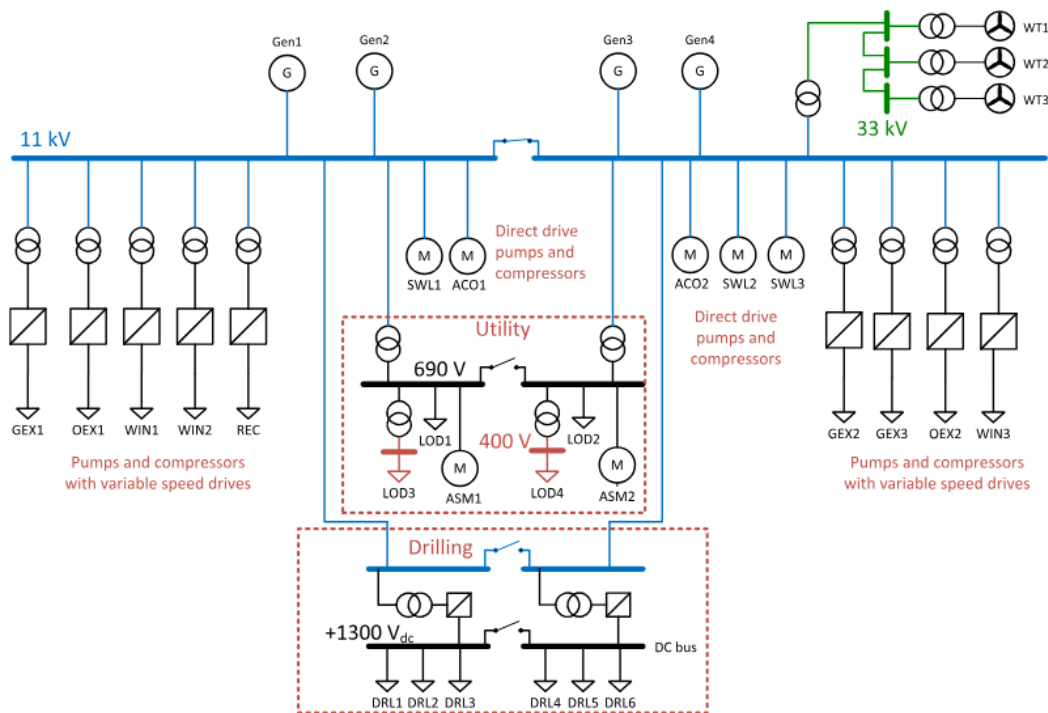


Figure 1: Caption

In the model as we use it, the 11 kV main bus is supplied by three 28 MVA synchronous (gas-turbine) generators, each with a governor, static exciter (AVR) and PSS, and it is the network slack. The large process drives (gas export/compression, water injection, oil export) appear on the grid as their converter DC-side loads plus LCL filter capacitors, and the smaller utility/accommodation machines

as aggregated loads. This gives a realistic, weak islanded grid with real inertia, real controllers and realistic short-circuit strength.

In this project we use LEOGO as a realistic host grid for a detailed wind-turbine model, rather than testing the turbine against an idealised infinite bus. As implemented in `test_WT_LEOGO_sim.py`, we replace LEOGO's original simplified WT1 converter (which only injected a fixed reference at Busbar WTG1 LV) with a physics-based turbine connected through a converter interface, adding aerodynamics, a two-mass drivetrain, pitch control, MPT tracking and optional frequency droop, plus optional tower structural modes (a lightly damped side-to-side mode near 0.234 Hz driven by generator torque, and a fore-aft mode near 0.235 Hz driven by rotor thrust) calibrated against a high-fidelity OpenFAST FMU. Because LEOGO is a small islanded system, a single turbine is electrically significant, which makes the coupling we care about visible: how platform disturbances (a generator trip, a main-bus load step, or a terminal fault) reach the turbine and excite the tower side-to-side mode through the generator torque, and how the converter and its droop respond to the resulting frequency and voltage swings. In short, LEOGO gives a credible offshore weak-grid context, modelled generators, inertia and controllers; in which to test how our OpenFAST-anchored turbine and tower model behaves under realistic events.

2 Outer-loop droop control

2.1 Implementation of droop control

New parameters:

- $f_{\text{nom_hz}}$, droop_enable , $K_{\text{droop_pu_per_hz}}$, headroom_pu

2.1.1 Control law

2.1.2 Frequency measurement

The droop controller uses a system-frequency signal calculated from the inertia-weighted centre-of-inertia (COI) speed of the three synchronous generators in the LEOGO model. The COI approach provides one common frequency signal for the interconnected system, rather than relying on the rotor speed of a single generator. The grid-frequency signal is defined as

$$f_{\text{grid}}(t) = f_{\text{nom}} + f_{\text{nom}} [\omega_{\text{COI}}(t) - \omega_{\text{COI},0}], \quad (1)$$

where $f_{\text{nom}} = 50$ Hz, ω_{COI} is the inertia-weighted per-unit rotational speed, and $\omega_{\text{COI},0}$ is its pre-disturbance value. Consequently, the controller reacts to the aggregate electromechanical response of the LEOGO system.

In a physical wind turbine, the frequency would normally be estimated locally at the point of connection, for example through a phase-locked loop or another synchronization mechanism. The COI frequency is used here as a system-level measurement in order to isolate and analyze the interaction between the wind turbine and the overall LEOGO frequency response.

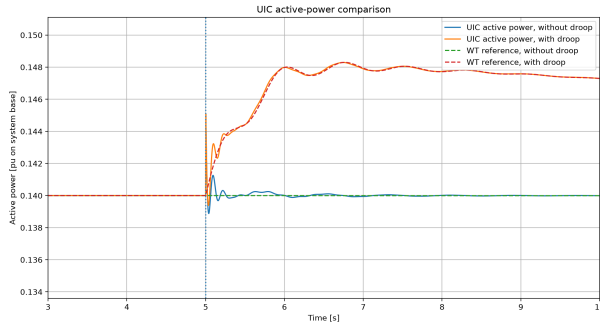
2.2 Droop tuning

2.3 Simulations and results

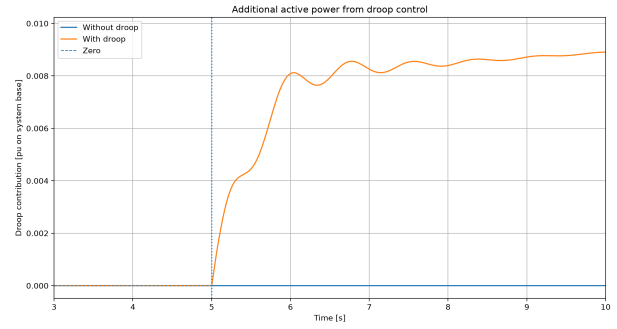
The droop controller in the baseline turbine model is tested against a single, well-defined grid event: a 5 MW active-power load step applied as a shunt on the 11 kV main bus. The step increases the platform demand, the LEOGO gas-turbine generators react, and the COI frequency falls. The same event is run twice — once with the droop disabled and once enabled — keeping wind speed, operating point, converter settings and event timing identical, so that any difference in the response is due to the droop alone.

2.3.1 Abrupt load-step

Here the full 5 MW is applied instantaneously. With the droop disabled the turbine holds its active-power reference constant and the gas turbines absorb the entire step, giving the deepest frequency dip. With the droop enabled the controller reads the falling COI frequency and raises the turbine active-power reference through the droop law, so the converter injects additional power that helps arrest the dip. Figure 2 compares the UIC active power for the two cases: with droop, the injected power steps up right after the event; without it, the power stays flat.

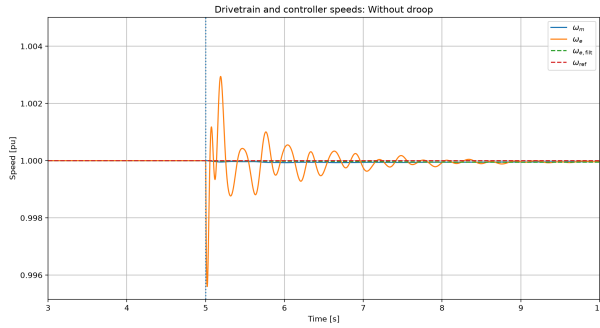


(a) UIC active power, abrupt load step.

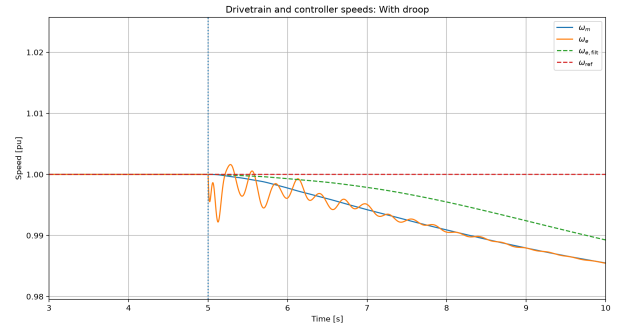


(b) Droop contribution, abrupt load step.

Figure 2: Active-power and droop response for the abrupt 5 MW load step.

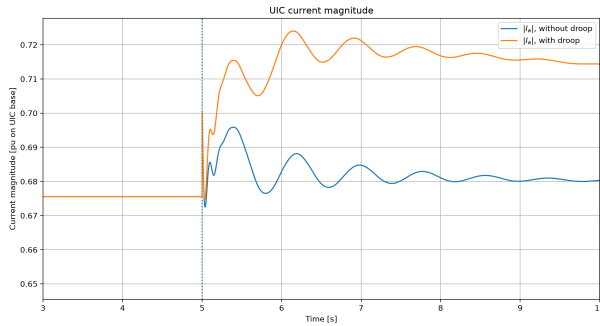


(a) Without droop control.

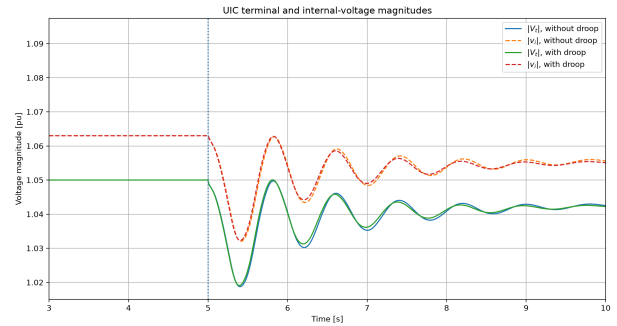


(b) With droop control.

Figure 3: Drivetrain-speed response following the abrupt 5 MW load step, comparing rotor-side and generator-side speed with and without droop control.



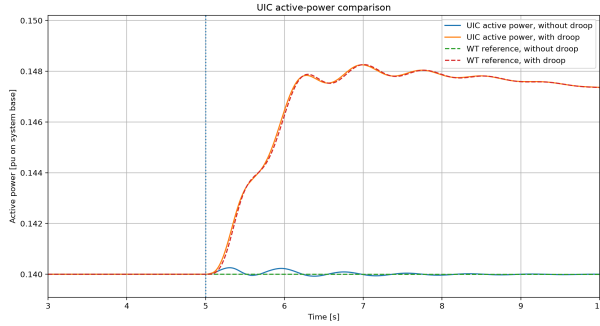
(a) UIC current magnitude.



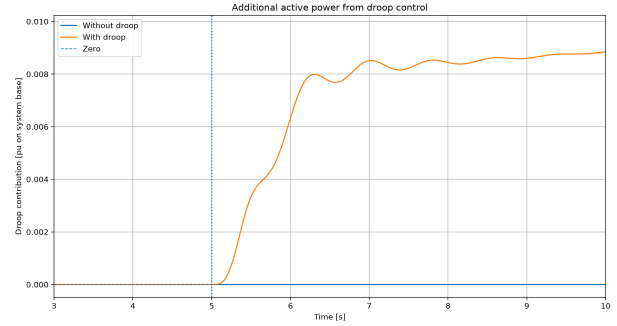
(b) UIC voltage magnitude.

Figure 4: Electrical response at the UIC terminal following the abrupt 5 MW load step, showing current-magnitude and voltage-magnitude transients for the no-droop and droop cases.

2.3.2 Smooth load-step

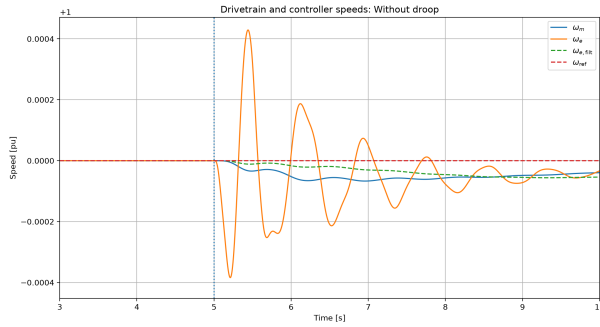


(a) UIC active power, smooth load step.

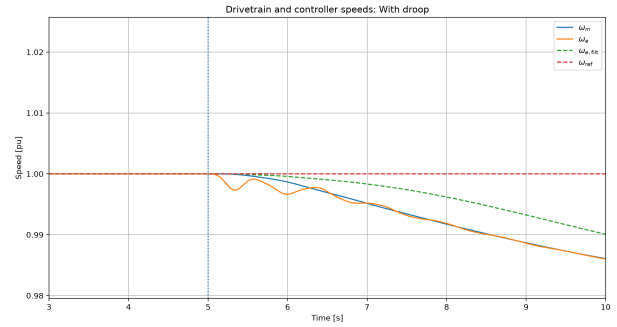


(b) Droop contribution, smooth load step.

Figure 5: Active-power and droop response for the smooth 5 MW load step.

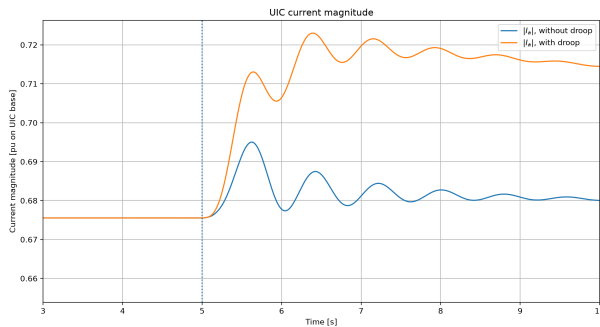


(a) Without droop control.

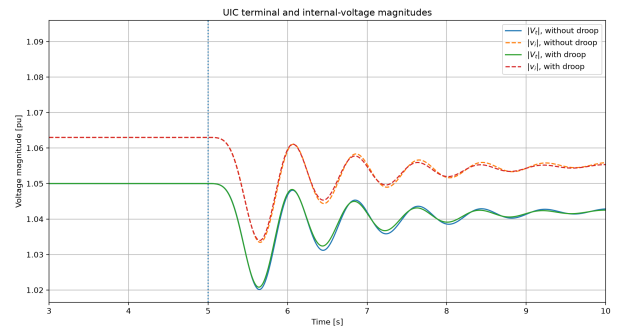


(b) With droop control.

Figure 6: Drivetrain-speed response following the smooth 5 MW load step, comparing rotor-side and generator-side speed with and without droop control.



(a) UIC current magnitude.



(b) UIC voltage magnitude.

Figure 7: UIC current and voltage response following the smooth 5 MW load step, comparing the no-droop and droop cases.

2.4 Discussion

The load-step simulations show that the droop controller provides the expected active-power support after the 5 MW disturbance. Without droop, the wind turbine keeps its active-power reference nearly constant, and the LEOGO gas-turbine generators must absorb most of the power imbalance. With droop enabled, the falling grid frequency increases the turbine power reference, and the UIC injects additional active power into the platform grid.

The difference between the abrupt and smooth load steps is mainly visible in the initial transient. The abrupt step excites stronger oscillations in the UIC power, drivetrain speeds, current and voltage, because the full disturbance is applied instantaneously. The smooth load step produces the same overall power imbalance, but the 0.5 s ramp reduces the sharp electrical excitation and gives a cleaner response.

The drivetrain-speed plots show that droop support is not only an electrical effect. When the turbine increases its active power, the electromagnetic torque changes and the generator-side speed is pulled down more clearly than in the no-droop case. This confirms that the frequency-support action creates an electrical-to-mechanical coupling path into the wind turbine drivetrain.

The UIC current and voltage responses also show the cost of providing support. The droop case gives a higher current magnitude and a more pronounced voltage response than the no-droop case, especially during the first seconds after the disturbance. Overall, the controller improves active-power support to the weak islanded grid, but it also increases the mechanical and electrical stress seen by the turbine and converter. This motivates the later investigation of whether such support can excite specific drivetrain or tower modes.

3 Expanding the baseline wind turbine model

The baseline model of the previous section captures the two-mass drivetrain, and with it the drivetrain torsional mode, but it stands on a rigid tower. To study whether a grid disturbance can reach the tower, we extend the model with the two first tower structural modes. This baseline is the natural model for that question: because its generator torque is tied directly to the electrical power, $T_e = P_e/(\omega_{e,\text{filt}} \eta)$, its electrical-to-mechanical path is open by construction. That is the opposite of the full-converter OpenFAST turbine, where the converter and the speed-controlled torque close the path, and it lets us sweep the interaction cheaply.

3.1 Adding the tower structural modes

We add two prescribed second-order modal degrees of freedom to the turbine: the tower side-to-side (SS) mode and the tower fore-aft (FA) mode. Each is a damped oscillator driven by the mechanical load that physically excites it. The side-to-side mode is driven by the generator torque, whose nacelle reaction moment rolls the nacelle laterally,

$$\ddot{q}_{ss} + 2\zeta_{ss}\omega_{ss}\dot{q}_{ss} + \omega_{ss}^2 q_{ss} = g_{ss} T_e, \quad (2)$$

and the fore-aft mode is driven by the rotor thrust,

$$\ddot{q}_{fa} + 2\zeta_{fa}\omega_{fa}\dot{q}_{fa} + \omega_{fa}^2 q_{fa} = g_{fa} F_T, \quad (3)$$

where $\omega = 2\pi f$ and g is a modal input gain. The frequencies, damping ratios and gains are prescribed and calibrated against the OpenFAST ring-down (Table 1), so the reduced model reproduces the same tower resonances as the high-fidelity model while remaining a small set of ordinary differential equations. The drivetrain torsional mode, by contrast, is not prescribed: it emerges from the two-mass drivetrain itself.

The key consequence of (2) is that the generator torque enters the side-to-side equation explicitly, and the generator torque follows the electrical power. A grid disturbance therefore reaches the side-to-side mode directly, with no control loop in between. The fore-aft mode is driven by thrust, for which the grid has no direct path.

Table 1: Structural modes added to the baseline turbine. The tower modes are prescribed and calibrated to the OpenFAST ring-down; the drivetrain torsion is emergent from the two-mass drivetrain.

Mode	f [Hz]	ζ [%]	Driven by
Tower side-to-side	0.234	0.34	generator torque T_e
Tower fore-aft	0.235	1.25	rotor thrust F_T
Drivetrain torsion	3.49	4.45	shaft twist (emergent)

3.2 Simulations and results

Running LEOGO operating scenarios

We drive the coupled LEOGO network with two representative platform disturbances, each applied as a cyclic real-power load at the main grid bus. The first is a slug-flow process-load pulsation, a

slow low-frequency oscillation typical of multiphase flow on an oil and gas platform. The second is a compressor pulse, a faster cyclic load from compressor operation. Both upset the power balance, so the LEOGO gas-turbine sets react and the grid frequency oscillates at the forcing frequency, which the turbine feels through T_e .

3.2.1 Shaft-torsion mode

Driving the grid load near the drivetrain torsional frequency (3.49 Hz) resonates the two-mass drivetrain: the oscillating electrical power modulates T_e , which enters the generator-mass equation of motion and builds up the shaft twist. Figure 8 shows the shaft response to a process pulsation at the torsional frequency.

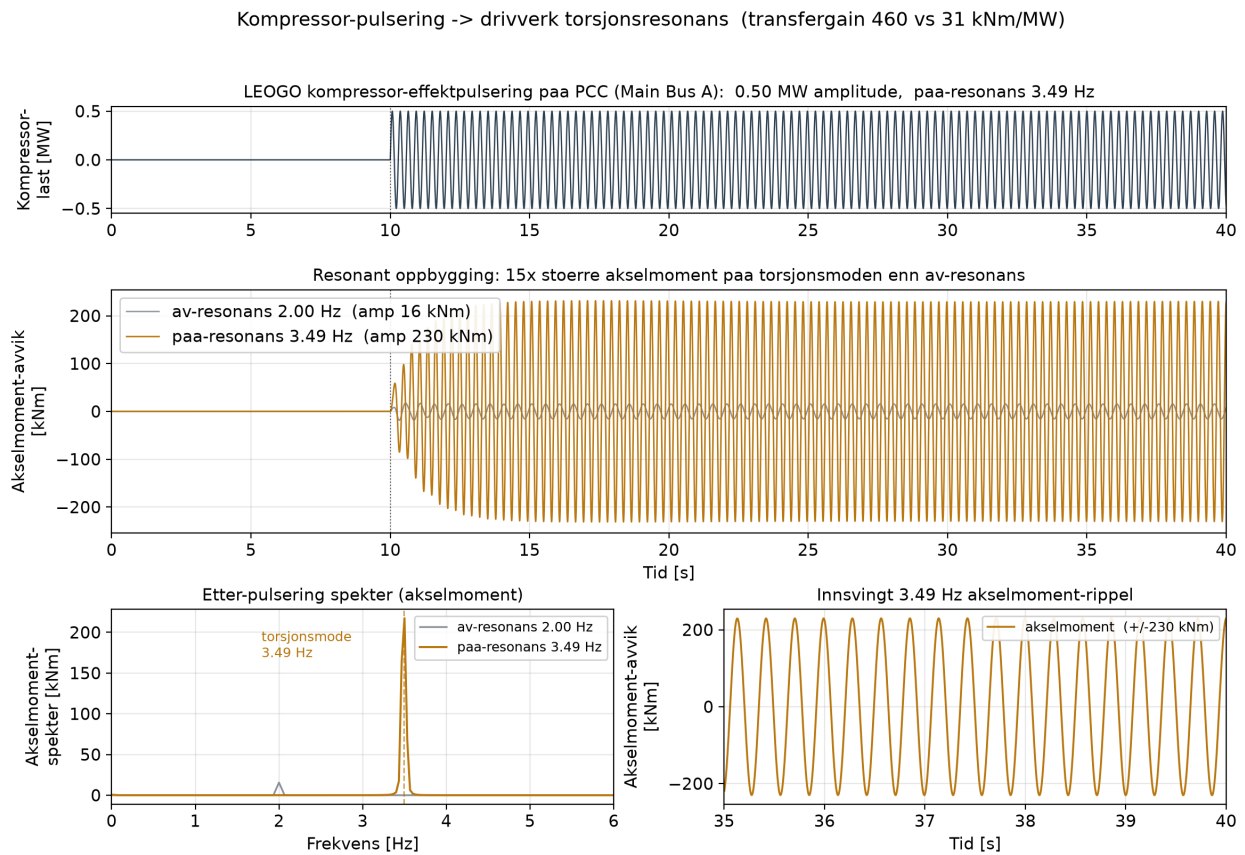


Figure 8: Baseline drivetrain response to a process-load pulsation near the torsional frequency (3.49 Hz): the shaft torque builds up as the electrical power drives the generator mass, confirming the grid-to-drivetrain resonance.

3.2.2 Tower fore-aft DOF

The fore-aft mode responds only weakly, on resonance as well as off. It is driven by rotor thrust, and the grid has no thrust channel. The one indirect route, $\text{grid} \rightarrow P_e \rightarrow T_e \rightarrow \omega_m \rightarrow \lambda \rightarrow C_T \rightarrow F_T$, is throttled by the large rotor inertia, which keeps the rotor speed and hence the thrust nearly constant at fixed wind. Comparing the on-resonance and off-resonance cases, the fore-aft amplitude stays essentially flat, so the grid does not reach the fore-aft mode.

3.2.3 Tower side-to-side DOF

The side-to-side mode is the one the grid can reach. A process pulsation at the side-to-side frequency (0.234 Hz) modulates T_e at that frequency and drives (2) on resonance, so the side-to-side amplitude grows sharply, while an off-resonance pulsation leaves it small. Figure 9 shows the side-to-side response to a slug-flow pulsation on resonance. The response also depends on the frequency-support droop, which reshapes how the electrical power, and therefore T_e , follows the grid frequency; Figure 10 shows the settled side-to-side amplitude as the droop gain is swept.

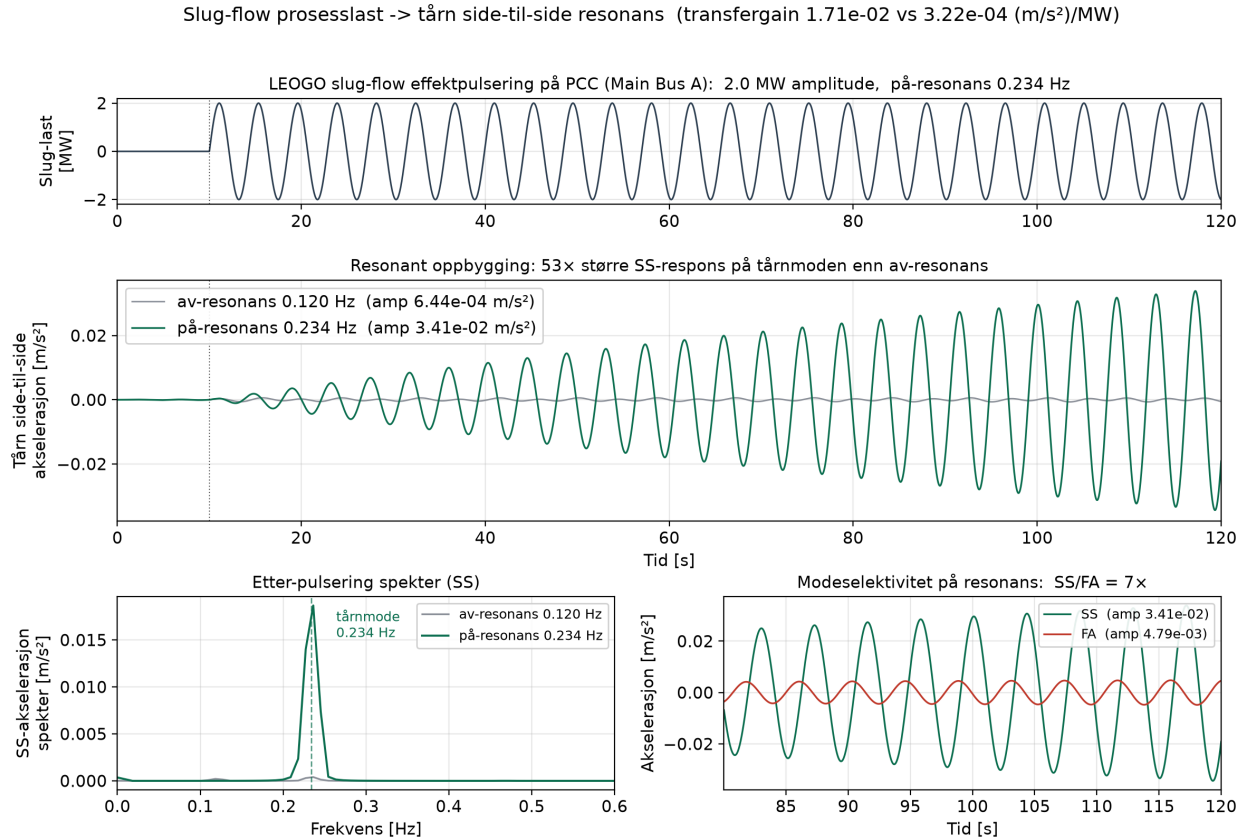


Figure 9: Baseline tower side-to-side response to a slug-flow process-load pulsation at the side-to-side frequency (0.234 Hz): the mode is driven through the open $T_e = P_e/(\omega_{e,\text{flt}}\eta)$ path and resonates, in contrast to the flat fore-aft response.

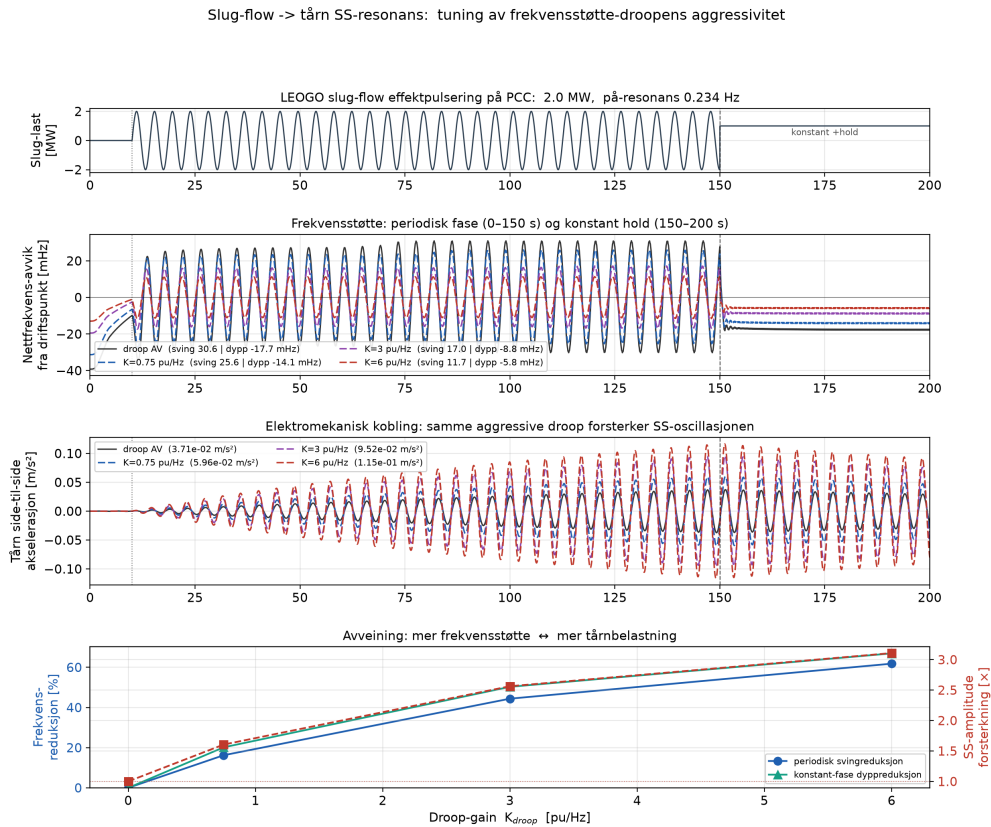


Figure 10: Settled side-to-side amplitude as the frequency-support droop gain is varied under the slug-flow pulsation, showing how the droop reshapes the electromechanical coupling into the tower.

3.3 Discussion

With the tower modes added, the baseline model shows a clear selectivity: a grid disturbance reaches the drivetrain torsional mode and the tower side-to-side mode, both driven by the generator torque, but not the fore-aft mode, driven by thrust. This is the same selectivity seen in the OpenFAST turbine, but the mechanism is the mirror image. In the baseline the torque path is open by construction, so the coupling exists without any control action. In the full-converter OpenFAST turbine the same path is closed by the converter, and the coupling appears only when a frequency-support loop reintroduces it. The two models therefore bracket the real turbine: a directly coupled machine, where the grid inherently reaches the structure, and a decoupled full-converter machine, where it does not unless the control opens the path.

One limitation is worth stating plainly. The tower modes here are prescribed and calibrated to the OpenFAST ring-down, so they reproduce the tower resonances rather than predict them; the frequency and damping are inputs, not outputs. The value of the reduced model is not a first-principles tower prediction but a fast, open-path platform on which the grid-to-tower interaction can be swept and understood.

4 Modal analysis

Before studying how a platform or grid disturbance excites the turbine, we identify the natural frequencies and damping ratios of its mechanical modes. The two turbine models are of a fundamentally different nature; one is a set of analytic equations, the other a compiled black box; so a complementary method is used for each. In both cases the turbine is connected to a stiff infinite bus instead of the LEOGO grid, which removes the network and gas-turbine modes and exposes the turbine's own structural and electromechanical modes.

4.1 Baseline model: eigenvalue analysis

The baseline model is a set of differential–algebraic equations. It is linearised about its operating point and the system matrix is eigen-decomposed; each complex-conjugate eigenvalue pair yields a mode's natural frequency $f = \text{Im}(\lambda)/2\pi$ and damping ratio $\zeta = -\text{Re}(\lambda)/|\lambda|$, and the participation factors show which states dominate it. The drivetrain torsional mode emerges directly from this analysis at 3.49 Hz with 4.45% damping, dominated by the generator-speed and shaft-twist states; Figure ?? shows the eigenvalue spectrum with this mode highlighted.

The two tower modes, by contrast, are not emergent. In the baseline model the tower side-to-side and fore-aft motions are represented by prescribed second-order oscillators whose frequency and damping are set as parameters (Table 2) and calibrated against the OpenFAST ring-down below. The drivetrain torsion is therefore the only mode the baseline model predicts independently.

Table 2: Modal parameters used in the simplified wind-turbine model. The tower modes are prescribed (calibrated to OpenFAST); the drivetrain torsion is computed from the eigenvalue analysis.

Mode	f [Hz]	ζ [%]
Tower side-to-side	0.234	0.34
Tower fore-aft	0.235	1.25
Drivetrain torsion	3.49	4.45

4.2 OpenFAST model: time-domain identification

The OpenFAST turbine is a compiled FMU and is effectively a black box to the power-system solver, so it cannot be linearised. Its modes are instead identified from the FMU's own transient response. With the structural degrees of freedom enabled, a short generator-torque impulse is applied at constant wind and the free ring-down of the tower-top accelerations (side-to-side, fore-aft) and the shaft torque is recorded. Each signal is band-pass filtered and detrended, and its natural frequency and damping ratio are extracted from an FFT peak and the log-decrement of the decay envelope.

This yields the lightly damped tower side-to-side and fore-aft modes near 0.23 Hz and the fast-decaying drivetrain torsional burst at 3.05 Hz (Table 3); Figure ?? shows the side-to-side ring-down and its spectrum together with the torsional burst.

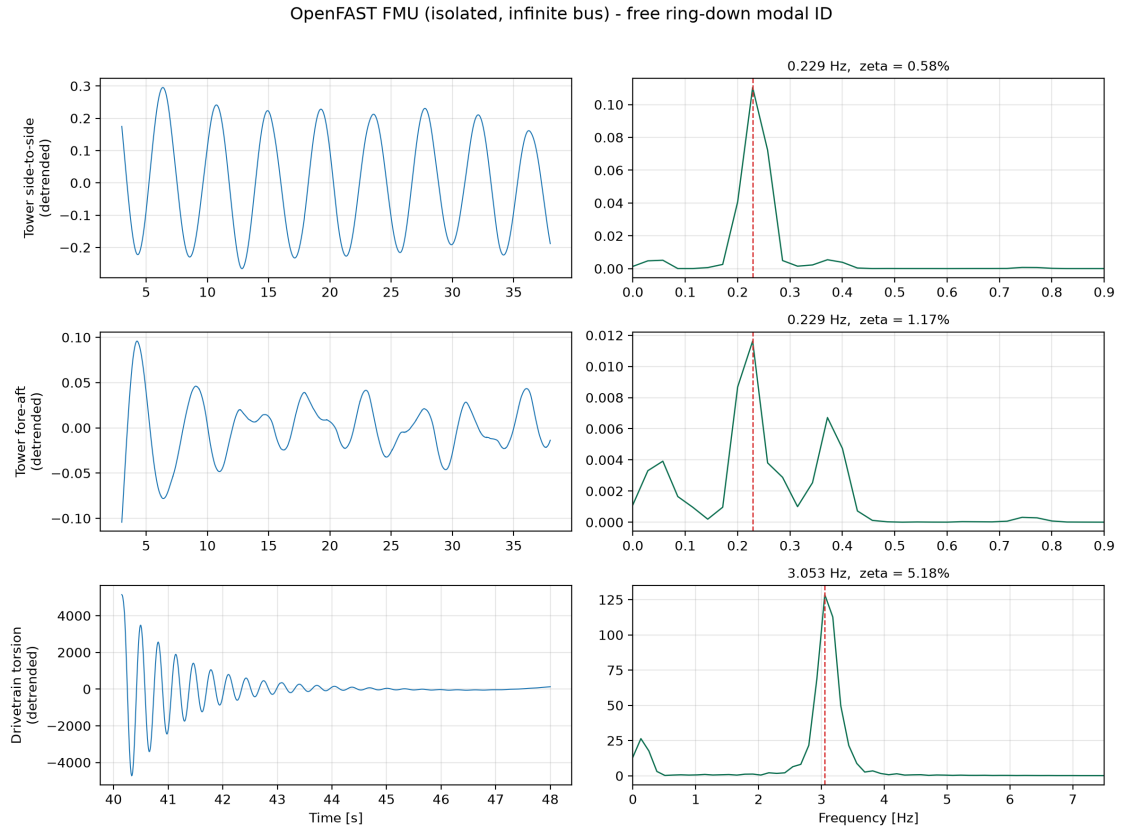


Figure 11: Caption

Table 3: Modal parameters identified from the OpenFAST wind-turbine model by time-domain ring-down.

Mode	f [Hz]	ζ [%]
Tower side-to-side	0.229	0.58
Tower fore-aft	0.229	1.17
Drivetrain torsion	3.05	5.18

4.3 Linking the two models

The two methods agree where they can be compared independently: the drivetrain torsional mode is predicted at 3.49 Hz by the baseline eigenvalue analysis and measured at 3.05 Hz in the OpenFAST ring-down, with consistent damping (4.45% vs 5.18%). The $\sim 13\%$ frequency offset reflects the baseline's lumped two-mass shaft being slightly stiffer than the distributed OpenFAST drivetrain. The baseline tower modes are not an independent check—their frequencies and damping are calibrated directly to the OpenFAST-identified values—but this calibration is exactly what lets the fast baseline model reproduce the same structural resonances as the high-fidelity model in the parameter sweeps that follow.

5 Implementing the ROSCO ZeroMQ torque channel

The ROSCO control mode (`VSContr1=5`) has full ownership of the generator torque, and every external FMI port that could carry a grid-driven torque into the turbine is either ignored (`GenSpd0rTrq`, `ElecPwrCom`) or crashes the compiled binary (`VSContr1=4`, see §??). That leaves only one channel through which an external signal can reach the actual generator torque without recompiling the FMU: ROSCO's own ZeroMQ interface, which is designed to accept run-time set-points from an external process. My goal was to open that channel, so that the co-simulation driver could impose a generator-torque offset on the turbine at every control step. This section describes the communication mechanism. What value to put on the channel is the subject of the next section.

5.1 The ROSCO side: a blocking request client

ROSCO is run with `ZMQ_Mode=1`. By default the ZeroMQ path only carries a yaw command, so the controller is patched at the `libDISCON` level to apply the first returned set-point as a generator-torque offset ΔT_{zmq} that is added to its internally computed demand,

$$T_{\text{gen}}^{\text{applied}} = T_{\text{gen}}^{\text{ROSCO}} + \Delta T_{\text{zmq}}, \quad (4)$$

where $T_{\text{gen}}^{\text{ROSCO}}$ is the ordinary $K\omega^2$ (Region 2) or constant-power (Region 3) torque. Once every `ZMQ_UpdatePeriod` = 0.01s, which is equal to the co-simulation macro step, and from inside `fmU.doStep`, ROSCO opens a ZeroMQ *request* (REQ) socket to `ZMQ_CommAddress = tcp://localhost:5555`. It sends a message of 17 turbine measurements, among them the simulation time, generator speed and measured generator torque, and then blocks until it receives 8 set-points in return.

5.2 The server: a reply responder in a daemon thread

The counterpart is a ZeroMQ *reply* (REP) server, `GridTorqueResponder`, bound to `tcp://*:5555`. It answers each request with the eight set-points $[\Delta T, 0, 0, 0, 0, 1, 1, 1]$, where the first entry is the torque offset in newton-metres and the rest are neutral pass-through values. The reply is null-terminated so that ROSCO's C `strtok` parser stops cleanly at the message end.

The awkward part is that ROSCO's request is issued from within `doStep` on the main simulation thread. If that same thread also had to service the socket, the simulation would deadlock, because it cannot be blocked inside `doStep` waiting for a reply and be the process that sends the reply at the same time. The REP socket is therefore serviced by a background daemon thread that does nothing but receive, reply and log. Each exchange takes microseconds, so it never stalls `doStep`.

5.3 Synchronisation with the co-simulation

The value returned on the channel is not computed in the socket thread. It is computed deterministically on the main thread, once per macro step. Each step the driver calls `update(t, ·)`, which stores a ready-to-send offset ΔT behind a lock before the FMU, and hence ROSCO's request, is advanced. The socket thread then simply returns whatever offset is current. Because `update` always runs before `step_fmU`, the offset ROSCO reads belongs to the current step, the computation is free of data races, and the coupling stays a clean once-per-step Jacobi exchange. The offset itself can be anything. A directly prescribed value can be placed on the channel, such as the open-loop torque impulse used to ring the drivetrain for the modal identification of §4, or it can be produced by a feedback law.

From an open channel to a closed loop

At this point the plumbing is complete. The ZeroMQ channel lets the driver impose an arbitrary generator-torque offset (4) on the OpenFAST turbine at every macro step, past the controller that would otherwise block it. A communication channel does not by itself close the electromechanical loop, though. It only carries whatever number we choose to send. To turn a live grid disturbance into a physically meaningful torque, one that makes the turbine respond to the network and feed power back into it, we need a control law that maps the measured grid frequency to the offset ΔT . That law is the droop and virtual-inertia frequency-support response, and it is the subject of the next section.

6 Adding droop and virtual inertia

The previous section opened the ROSCO torque channel but left it carrying an arbitrary number. This section gives that number a control law and uses it to close the electromechanical loop in both directions: first the forward path, where a generator-torque modulation reaches the grid, and then the return path, where a grid disturbance drives the turbine.

6.1 The forward path: torque to grid

The mechanical-to-electrical direction is open. Placing a small sinusoidal torque offset on the ZeroMQ channel makes the generator-torque swing propagate all the way into the islanded grid,

$$\Delta T \rightarrow P_{\text{gen}} \rightarrow \text{UIC injects the modulated power at the PCC} \rightarrow f_{\text{grid}}.$$

The applied torque changes the generator power, the UIC converter injects it at the point of common coupling, and the resulting power imbalance moves the grid frequency, to which the LEOGO gas-turbine sets react. This confirms the turbine can drive the network through the channel of §5. What is still missing is the reverse: a way for the network to drive the turbine.

6.2 The frequency-support control law

For a grid event to reach the turbine, the ZeroMQ offset must be computed from the measured grid frequency. The frequency-support law has two active terms, a droop term proportional to the frequency deviation and a synthetic-inertia term proportional to its rate of change, plus a standing de-load term whose role is explained below:

$$\Delta T = -T_{\text{deload}} - K_{\text{droop}}(\tilde{f}_{\text{grid}} - f_{\text{nom}}) - K_{\text{inertia}} \frac{d\tilde{f}_{\text{grid}}}{dt} \quad [\text{N m}]. \quad (5)$$

Here $\tilde{f}_{\text{grid}}$ is the centre-of-inertia grid frequency after a low-pass filter (discussed at the end of this section). The droop reacts to the deviation and so holds the frequency up over time; the inertia term reacts to the rate of change and so slows the initial fall, drawing a short burst of power from the rotor's kinetic energy.

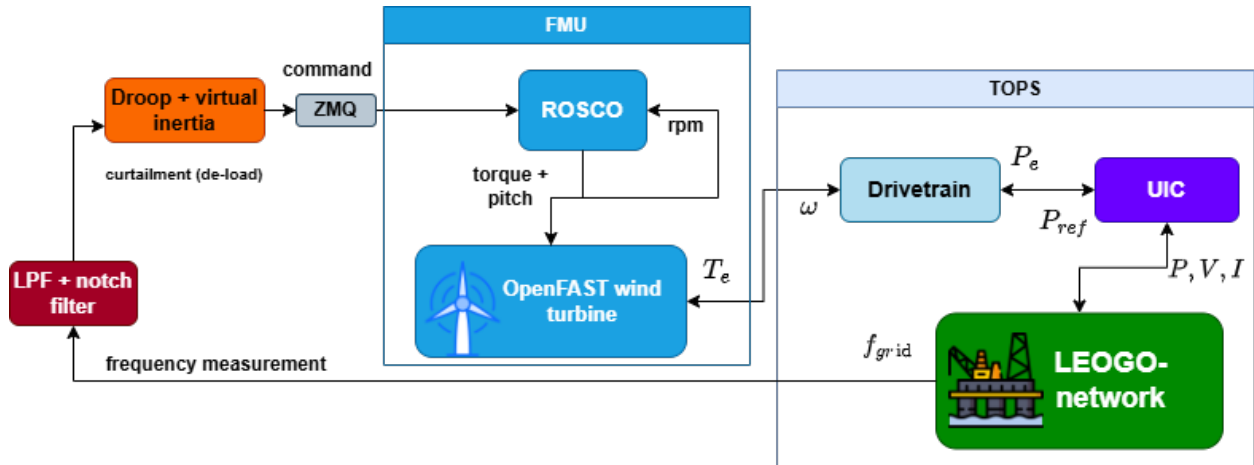


Figure 12: Co-simulation setup coupling the LEOGO platform grid, TOPS/UIC model and OpenFAST FMU. The frequency-support loop modifies the turbine torque command from the measured grid frequency, creating a coupling path between grid disturbances and turbine mechanical dynamics.

6.3 Disturbances

Two disturbances are used. The first is a cyclic process load: a shunt admittance at the main grid bus draws a real power scaled by a zero-mean sinusoid, giving a ± 3 MW oscillation at 0.234 Hz. It is zero-mean so it excites the grid without shifting the operating point, and the frequency is chosen to match the tower side-to-side mode, motivated by slow cyclic phenomena such as slug-flow load variations or pump and compressor pulsations.

The second is a gas-turbine trip. LEOGO has three 28 M gas turbines dispatched near 15.8 MW each, so the design single contingency is the loss of one unit, a sustained 15.8 MW generation deficit. This is applied as a load step at the main grid bus and drives the frequency to a nadir, which is the natural test of frequency support.

6.4 Reserve: what each term needs

In Region 3 the turbine runs near its rating, so it has no upward power headroom. A droop that raises power then simply pushes the machine past its 15 MW rating, which is not physical for sustained support. The two terms need different kinds of reserve. The droop is sustained, so it needs a matched *power* reserve, created by de-loading (curtailing) the turbine with $T_{\text{de-load}}$. The synthetic inertia is only transient, so it needs no standing reserve: its brief burst comes from rotor kinetic energy and from the converter's short-term overload capability, after which the power returns to its de-loaded level.

6.5 The simulation

We apply the gas-turbine trip to the coupled LEOGO and OpenFAST system and compare three cases at a de-loaded operating point (the turbine curtailed to about 8 MW, roughly half its rating, to create reserve): no support, droop only, and droop with synthetic inertia. Figure 13 shows the grid frequency and the turbine power.

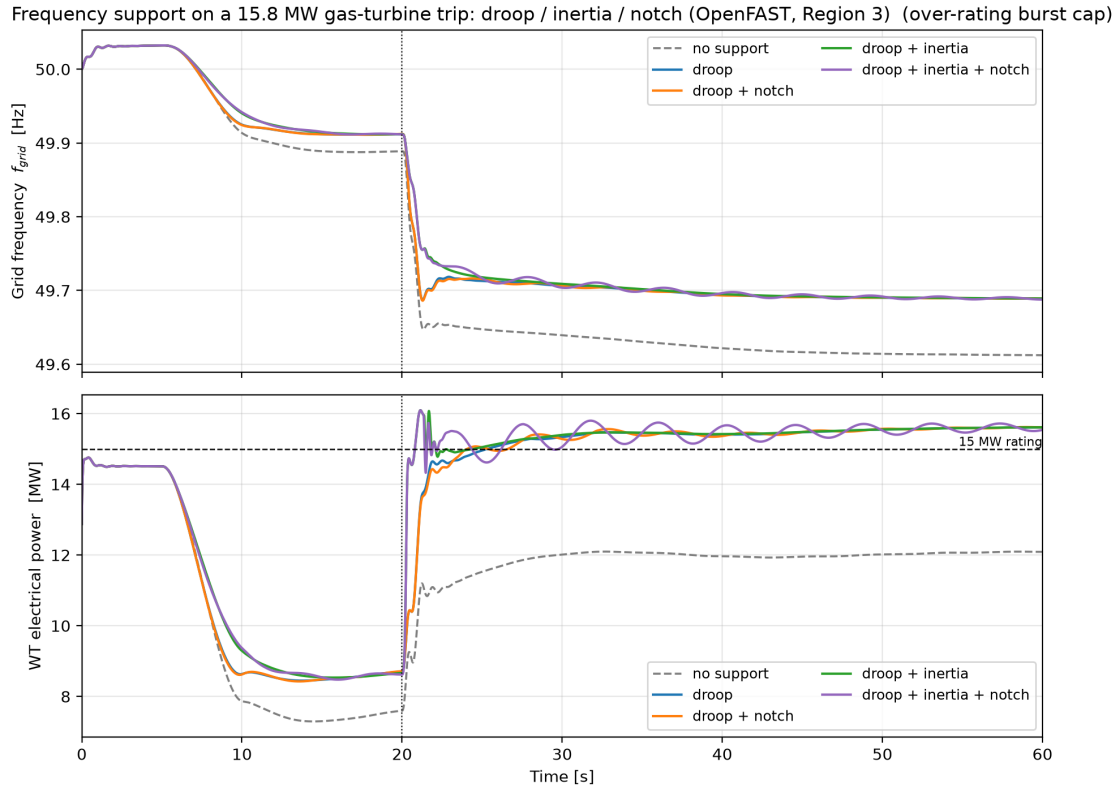


Figure 13: Grid-frequency and turbine-power response following a 15.8 MW gas-turbine trip, comparing no support, droop control, droop with synthetic inertia, and notch-filtered support cases.

The two terms act on different parts of the response, exactly as (5) predicts (Table 4). The droop lifts the frequency nadir by about 19 % (from 49.72 Hz to 49.78 Hz) but leaves the initial rate of change of frequency (RoCoF) essentially untouched, since it does not react to the rate. The synthetic inertia lowers the RoCoF by about 40 % (from 0.37 Hz s^{-1} to 0.22 Hz s^{-1}) and adds little to the nadir. The two are therefore complementary: inertia for the fast rate, droop for the depth and the settled deviation.

The power panel confirms that the support stays physical. With the de-load matched to the droop, the sustained droop response settles near the 15 MW rating rather than past it, while the synthetic-inertia burst rises briefly to about 16 MW and then falls back, using the transient overload margin as intended.

Table 4: Frequency support on a 15.8 MW gas-turbine trip (de-loaded OpenFAST turbine, Region 3).

Case	RoCoF [Hz s^{-1}]	Nadir [Hz]	Settled power [MW]
No support	0.37	49.72	12.1
Droop	0.37	49.78	15.6
Droop + inertia	0.22	49.78	15.6

Figure 14 shows that the frequency-support action also affects the turbine mechanically. When the turbine is de-loaded, the rotor speed decreases before the trip as power is curtailed to create reserve. After the gas-turbine trip, the droop cases increase generator torque in order to deliver additional electrical power, which is reflected in both the generator-torque and shaft-torque responses. The synthetic-inertia case gives a sharper transient torque contribution immediately after the trip, while the droop-only case mainly affects the slower recovery.

The pitch response follows the change in operating point and helps regulate the turbine as the support action develops. The key observation is that frequency support is not only an electrical grid service: it changes the torque applied to the drivetrain and therefore modifies rotor speed, shaft torque and pitch behaviour. This confirms that the support loop creates an electromechanical coupling path from the LEOGO frequency disturbance into the OpenFAST turbine dynamics.

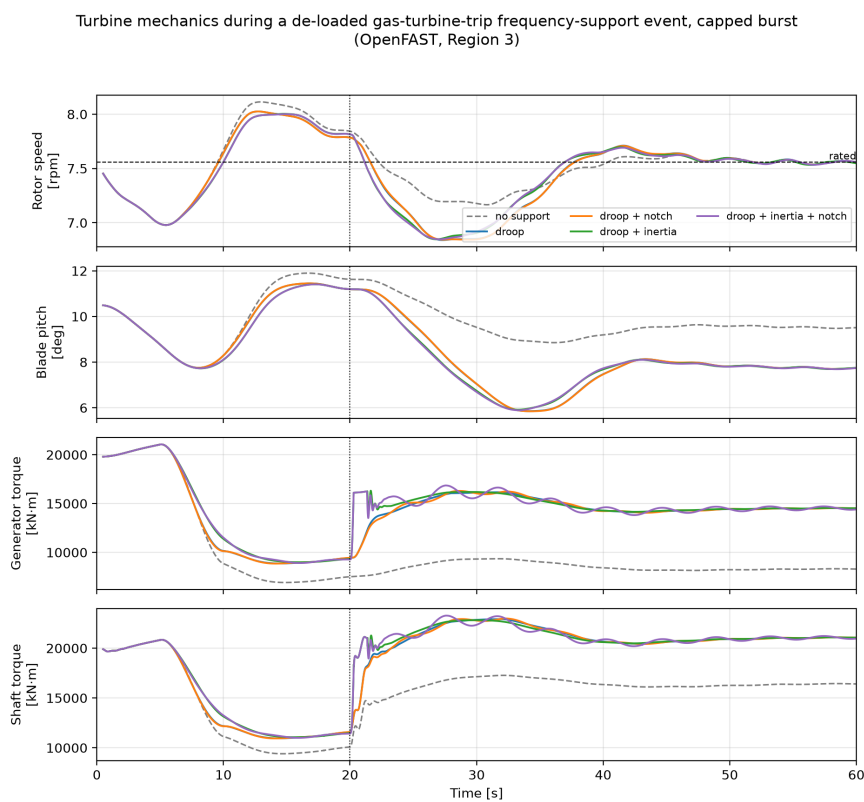


Figure 14: Mechanical response of the de-loaded OpenFAST turbine during the 15.8 MW gas-turbine trip, showing rotor speed, pitch angle, generator torque, and shaft torque for the different frequency-support cases.

6.6 The filter trade-off

The synthetic inertia only bites when the fast rate of change survives to the control law, so the low-pass filter on f_{grid} matters. A low corner frequency (here 0.5 Hz) rejects the lightly damped ~ 5.3 Hz LEOGO electrical mode, which a high-gain droop reading the raw frequency would otherwise pump into the generator torque; but the same smoothing removes the fast df/dt , so the inertia barely responds and the RoCoF improvement collapses to a few percent. Opening the corner to 2 Hz lets the rate through and restores the 40 % RoCoF reduction, at the cost of admitting more fast grid dynamics into the torque. The filter corner is thus a genuine trade-off between mode rejection and inertial responsiveness, and it must be tuned for the grid at hand.

6.7 Adding a notch-filter

The droop and synthetic-inertia terms act on the measured grid frequency, so any frequency content they pass is written straight onto the generator torque. When a disturbance sits at the tower side-to-side frequency, the support then modulates the torque at exactly that frequency and pumps the tower resonance. A notch filter removes this pumping without weakening the support. It is a second-order band-stop (an RBJ biquad in Direct Form I) placed on the frequency-support signal after the low-pass filter and tuned to the tower mode near 0.233 Hz. Its gain is unity at DC and at high frequency, so the slow droop and inertia response is left untouched, and it forms a deep, narrow null at the tower frequency, so the support no longer forces the tower there. The width of the null is set by the quality factor Q , with bandwidth roughly f_0/Q .

6.8 Grid excitation of the tower modes in the full model

In the reduced model of the generator-torque path was open by construction, so a grid disturbance reached the tower directly. In the full-converter OpenFAST turbine that path is closed by ROSCO, and a grid disturbance does not reach the tower on its own. The frequency-support loop of the previous sections deliberately reopens it, because a grid-frequency deviation is turned into a generator-torque offset, which is the same channel that drives the side-to-side mode. In this section we drive the coupled LEOGO and OpenFAST system with a cyclic platform load and ask two things. First, does the reopened path now let the grid excite the tower modes. Second, can the notch keep the frequency support while closing that path again for the tower.

6.8.1 Side-to-side mode

Figure 15 compares three cases at inertia 5e6: no support, droop with inertia, and droop with inertia and notch. The left panel sweeps the process-load frequency and plots the settled tower-top side-to-side acceleration. With no support the tower stays flat across the whole sweep, confirming that the closed path keeps the grid from reaching it. With droop and inertia a sharp resonance appears at the side-to-side frequency near 0.233 Hz, peaking at about 0.56 m/s^2 , so the support has reopened the electrical-to-mechanical path and the grid now drives the tower on resonance. Adding the notch collapses this peak to a small residual. The right panel confirms the same result in the time domain under a sustained 0.233 Hz platform load: the droop-and-inertia case settles into a large side-to-side oscillation of about 0.6 m/s^2 amplitude, while both the no-support and the notch cases stay essentially flat. The notch therefore cancels the tower excitation that the support would otherwise introduce, while leaving the frequency-support action itself intact (Section 6.7).

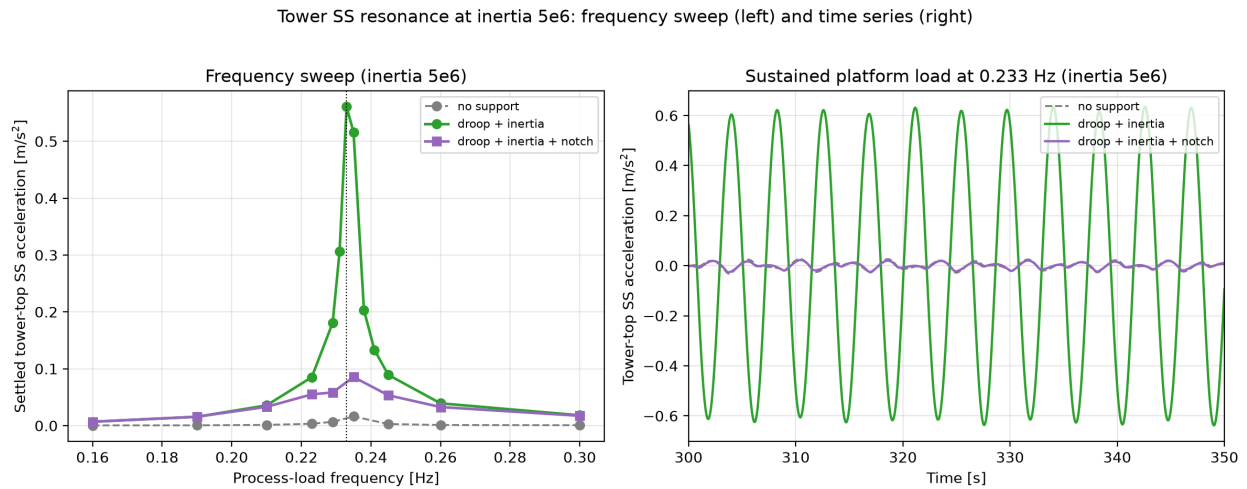


Figure 15: Tower side-to-side response to cyclic platform-load excitation. Droop with synthetic inertia excites the tower resonance, while the notch filter suppresses the response.

6.8.2 Fore-aft mode

Future work

7 Appendix

7.1 Hvordan er systemet vårt koblet til LEOGO?

7.1.1 LEOGO-modellen

LEOGO-nettet lastes som én datastruktur som TOPS bygger en `PowerSystemModel` av. Det inneholder 47 busser på flere spenningsnivå (33 kV vindsamling, 11 kV hovedbuss, 0.69 kV utility, 0.4 kV bolig og 3.3 kV VSD-lavsider), tre 28 MVA synkrongeneratorer (≈ 15.8 MW, $H \approx 7.0$) på `Main Bus A` med TGOV1-guvernører, SEXS-magnetisering og STAB1-stabilisatorer, prosess- og utility-lastene, samt LCL-filterkondensatorene. `Main Bus A` er slakk-bussen, så gassturbinene setter systemfrekvensen. I sin opprinnelige form er vindturbinen på `Busbar WTG1 LV` én forenklet omformer (`GridSideConverter_PV`) som bare injiserer en fast referanse.

7.1.2 Tilkoblingspunktet

Turbinen kobles inn på `Busbar WTG1 LV` (0.69 kV). Derfra går den elektriske veien opp til gassturbinene uendret:

`Busbar WTG1 LV` $\rightarrow$ `Trafo WTG1` $\rightarrow$ `Busbar WTG1 HV` $\rightarrow$ `Cable WTG1` $\rightarrow$ `Terminal WindPark` $\rightarrow$ `Trafo WindPark` $\rightarrow$ `Main Bus A`.

Turbinen samvirker altså med generatorene, guvernørene og magnetiseringen på `Main Bus A` gjennom denne trafo- og kabelveien.

7.1.3 Utbygging av den originale omformereren

Koblingen bygges i `build_model()`:

1. Ta en dyp kopi av hele LEOGO-nettet.
2. Fjern den originale `GridSideConverter_PV` (den forenklede WT1-omformereren).
3. Legg til `UIC_sig`-omformereren (navn `WT1_LEOGO`, 20 MVA, 0.69 kV) på *samme* buss.
4. Legg til OpenFAST-turbinen `FMUtoUICdrivetrain` med `UIC = WT1_LEOGO`.

Siden samme buss og tilkoblingspunkt gjenbrukes, står resten av LEOGO urørt; bare det som ligger bak `Busbar WTG1 LV` endres.

7.1.4 Grensesnittet turbin-nett

Turbinen kobles til omformereren gjennom tre standard TOPS-koblinger: den elektriske effekten `P_e` og omformerbasen `S_n_UIC` inn, og effektreferansen `P_ref` ut. Den aeroelastiske dynamikken lever i OpenFAST-FMU-en, mens `UIC_sig`-omformereren er den som faktisk injiserer strøm på `Busbar WTG1 LV`. Nettfrekvensen tas som den tregheitsvektede senter-av-inerti-hastigheten til de tre synkrongeneratorene, og når ZeroMQ-koblingen er aktiv driver denne frekvensen ROSCO-momentoffsetet, slik at den elektromekaniske sløyfen lukkes.

7.1.5 Hvorfor én enkelt turbin betyr noe

LEOGO er et lite øydrift-nett (topplast ≈ 47 MW) med gassturbinene som slakk. Én 15 MVA-turbin er derfor elektrisk signifikant, så en plattformforstyrrelse (et generatorutfall, et laststeg eller en feil) flytter frekvensen merkbart og når turbinen. Dette plasserer den høyoppløste turbinen i et realistisk, svakt øydrift-nett i stedet for mot en idealisert stiv buss.

7.1.6 LEOGO-spesifikke justeringer

Tre valgfrie justeringer gjøres på LEOGO-dataene før en kjøring:

- `--fix-leogo-xqt` retter et dataartefakt i generatorene ($X'_q < X''_q$, ikke-fysisk) som ellers gir en lett dempet ~ 5.3 Hz-mode.
- `--perfect-tracking` setter omformerens `perfect_tracking`-flagg (0: nettets frekvensavvik når turbinen; 1: omformeren holder sin interne frekvens fast og kobler den fra).
- `--wt-pref-mw` setter vindturbin-injeksjonen som lastflyten bruker når gassturbinene dispatches, slik at stasjonær nettfrekvens initialiseres på 50 Hz.

7.2 Endringer i windturbine.py

Den nye filen (windturbine_tower.py) erstatter windturbine.py, i dyn_models i src tops_openfast. Hele drivverket, aerodynamikken, pitch-regulatoren, MPT/Cp-håndteringen og UIC-grensesnittet er beholdt, og oppå dette er det lagt til:

- to strukturelle tårnmoder (side-til-side og fore-aft)
- droop-basert frekvensstøtte
- en valgfri toveiskobling mellom tårn og drivverk
- hjelpefunksjoner for kalibrering mot OpenFAST-FMU-en.

Med `ss_enable=0`, `fa_enable=0` og `droop_enable=0` oppfører modellen seg i praksis som den gamle `WindTurbine`.

Nye tilstander

`state_list` er utvidet fra 10 til 14 tilstander. De nye er:

- `q_ss`, `q_ss_dot`: tårnets side-til-side forskyvning og hastighet. `q_fa`, `q_da_dot`: tårnets fore-aft forskyvning og hastighet.

Når modene er avslått er de nye tilstande inerte (deriverte settes til null), så løsning er identisk med den gamle (for utenom droop).

7.2.1 Nye parametre

Parametermalen i docstringen har fått nye kolonner:

- **Droop:** `f_nom_hz`, `droop_enable`, `K_droop_pu_per_hz`, `headroom_pu`
- **Side-til-side:** `ss_enable`, `f_ss_hz`, `zeta_ss`, `g_ss`
- **Fore-aft:** `fa_enable`, `f_fa_hz`, `zeta_fa`, `g_fa`.

Alle leses via en ny hjelpefunksjon `_get_par(name, default)` som faller tilbake på fornuftige standardverdier hvis feltet mangler, slik at gamle parametermaler fortsatt lar seg konstruere. Eksempelverdiene i malen er også oppdatert til **IEA 15 MW** (bl.a. `S_n=15`, `f_ss=0.234`, `zeta_ss=0.0034`, `g_ss=0.01825`, `f_fa_hz=0.235`, `zeta_fa=0.0125`, `g_fa=0.102`).

7.2.2 Tårnets side-til-side-mode (SS)

- Ny modal oscillator i `state_derivatives`, drevet av generatormomentet:

$$\ddot{q}_{ss} + 2\zeta_{ss} \omega_{ss} \dot{q}_{ss} + \omega_{ss}^2 q_{ss} = g_{ss} T_e. \quad (6)$$

- Ny metode `ss_acceleration(x, v)` som gir tårntopp-akselerasjonen (analog til OpenFAST-utgangen `YawBrTAyp`).
- Ny metode `_electromagnetic_torque(x, v)` som beregner T_e frittstående (speiler formelen i `state_derivatives`) og brukes av SS-akselerasjonen.

- Initialisering: q_{ss} settes til statisk likevekt

$$q_{ss,0} = \frac{g_{ss} T_{e0}}{\omega_{ss}^2}, \quad (7)$$

slik at et konstant moment ikke gir en falsk oppstartstransient.

7.2.3 Tårnets fore-aft-mode (FA)

- Ny modal oscillator drevet av rotor-thrust:

$$\ddot{q}_{fa} + 2\zeta_{fa}\omega_{fa}\dot{q}_{fa} + \omega_{fa}^2 q_{fa} = g_{fa}\hat{F}_T. \quad (8)$$

- Nye metoder:

- `_thrust_ct_v2(x, v)` — thrust-proxy $C_T(\lambda, \beta) v^2$.
- `_fa_forcing(x, v)` — normalisert thrust-forcing ($\hat{F}_T = 1$ i likevekt).
- `fa_acceleration(x, v)` — fore-aft akselerasjon (analog til `YawBrTExp`).
- `load_and_set_Ct(x, v)` — laster thrust-koeffisienttabellen $C_T(\text{pitch}, \text{TSR})$ fra ROSCO-fila (helt ny; den gamle lastet bare C_p).

- Initialisering: lagrer likevekts-thrusten `_fa_thrust0` og setter

$$q_{fa,0} = \frac{g_{fa}}{\omega_{fa}^2}. \quad (9)$$

7.2.4 Valgfri toveiskobling $\leftrightarrow$ drivverk

Standard er foroverkoblet (`c=0`): tårnet drives av drivverket, men virker ikke tilbake, så elektrisk/drivver-løsningen er uendret og gamle løsningen er uendret og gamle resultater reproduseres bit-for-bit. Slås koblingen på, knyttes generatorrotor- og tårn-SS-akselerasjonen gjennom en symmetrisk positivt definit 2x2 massematrise. Da reagerer tårnbevegelsen tilbake på rotoren, T_e og nettet.

7.2.5 Droop / frekvensstøtte

- Ny `set_grid_frequency_hz(f)` for å mate inn den målte nettfrekvensen f_{grid} .
- Ny `_droop_command(p_available)` som implementerer droop-loven med de-loading (`headroom_pu`) og klipping mot tilgjengelig effekt:

$$P_{base} = \text{clip}(P_{avail} - P_{headroom}, 0, P_{avail}), \quad (10)$$

$$\Delta P = K_{droop} (f_{nom} - f_{grid}), \quad (11)$$

$$P_{ref} = \text{clip}(P_{base} + \Delta P, 0, P_{avail}). \quad (12)$$

- Ny `P_ref_components(x, v)` som returnerer MPT-grensen P_{avail} , basisreferansen P_{base} , droop-bidraget ΔP og den totale referansen P_{ref} .
- **P_ref er omskrevet:** den gamle versjonen returnerte ren MPT-effekt, mens den nye går via `P_ref_components / _droop_command`.

- `P_ref_from_wind` klipper nå mot ratet effekt og legger på droop-kommandoen (den gamle returnerte kun MPT-effekt).

7.2.6 Endringer i pitch-regulatoren (viktig atferdsendring)

I `state_derivatives`:

- **Gammel:** pitch-feilen brukte den *rå* generatorhastigheten,

$$e_{\omega} = \omega_e - \omega_{ref},$$

og region ble valgt med betingelsen `if e_omega < -0.001`.

- **Ny:** pitch-feilen bruker den *filtrerte* generatorhastigheten,

$$e_{\omega} = \omega_{e, \text{filt}} - \omega_{ref},$$

og region velges på *vindhastighet* (`use_region_2: v_{wind} < v_{rated}`).

7.2.7 Ny reset-funksjon

Ny `reset_tower_modes(x, v)` som nullstiller tårn-modenes tilstander til gjeldende statiske likevekt (bruker den innkjørte, settlede T_e og thrusten). Nyttig for å starte en «warmed-up» baseline uten residual ring-down.

7.2.8 Kalibrerings- og testhjelpere

Helt nytt for sveip mot FMU-en:

- `set_ss_test_modulation / _ss_test_factor` — legger en brøkvis momentmodulasjon $T_e (1 + m(t))$ på SS-forcingen (reproduserer FMU-sveipeprotokollen).
- `set_fa_test_modulation / _fa_test_factor` — tilsvarende for FA-thrusten.

Disse påvirker ikke de elektriske / drivverk-tilstandene (modene er foroverkoblet).

7.2.9 Vind

- **Gammel:** `wind_speed(x, v)` brukte tidsinterpolert vind (`_wind_interp(t)`), og `wind_speed_init` returnerte 14.0 m/s.
- **Ny:** returnerer **konstant** 11.0 m/s (både `wind_speed` og `wind_speed_init`). Vindfila lastes fortsatt, men brukes ikke i selve `wind_speed`. Konstant vind trengs for tårn-kalibreringen og ring-down-protokollen.

7.2.10 Diverse

- Klassen er **omdøpt** `WindTurbine` → `WindTurbineTower`, og en ny **alias-subklasse** `WindTurbineTower2` er lagt til (for en andre, uavhengig turbin på samme nett, spiller `WindTurbine2`).
- Stor ny **modul-docstring** med teori, kilder (Bossanyi; Burton et al.; Jonkman / FAST; Gaertner, IEA 15 MW; Kühn; Salzmann & van der Tempel) og kalibreringsutledninger.
- Import ryddet: `import os` fjernet (kun `pathlib.Path` brukes nå).

7.3 Ny FMU-simulator sammenlignet med den gamle

Den gamle driveren (`test_WT_FMU_drivetrain_sim.py`) kjører FMU-turbinen på et lite testnett for å verifisere drivverks- og momentkoblingen isolert. Den nye (`test_WT_LEOGO_FMU_sim.py`) setter den samme FMU-en inn i hele LEOGO-nettet og lukker den elektromekaniske sløyfen gjennom ROSCO. De viktigste forskjellene:

- **Nett.** Gammel: enkelt testnett (stiv buss). Ny: hele LEOGO, der `build_model()` bytter ut den originale WTG1-omformerer med `UIC_sig + FMUtoUICdrivetrain` på «Busbar WTG1 LV».
- **Struktur.** Gammel: ett flatt skript. Ny: delt i funksjoner (`build_model`, `grid_frequency_hz`, `collect_results`, `main`).
- **Konfigurasjon.** Gammel: miljøvariabler og hardkodede verdier. Ny: full `argparse`-CLI (`last`, `feil`, FMU-valg og hele frekvensstøtte-blokken).
- **Frekvensstøtte (nytt).** Ny beregner en treghetsvektet COI-frekvens og starter en ZeroMQ-server (`GridTorqueResponder`) som hvert steg svarer ROSCO med et momentoffset fra droop, syntetisk inert, LPF og notch. Gammel hadde bare en valgfri åpen-sløyfe strøminjeksjon.
- **Forstyrrelser.** Ny bruker et mykt laststeg (med ramper og valgfri sinus) på hovednett-bussen, pluss valgfri trefasefeil ved WT-bussen; forstyrrelsen ligger der frekvenshendelsen faktisk oppstår.
- **Co-simuleringsrekkefølge.** Ny mater nettfrekvensen til responderen *før* `step_fmu` (ROSCO sender sin blokkerende ZMQ-forespørsel inne i steget), og terminerer FMU-en *før* responderen lukkes (ROSCO sender en siste forespørsel under finalisering).
- **Resultat.** Gammel: omfattende moment-diagnostikk til en fast CSV med «thesis»-plott. Ny: én ryddig `collect_results()` med nett-signaler og de relevante FMU-utgangene (`YawBrTApp`, `YawBrTAxp`, `HSShftTq`, ...), skrevet til `--out`.
- **FMU-valg.** Ny velger bevisst mellom `fast_debug.fmu` (side-til-side `YawBrTApp`) og `fast.fmu` (`HSShftTq`, raskere) via `--fmu`.

7.4 Endringer i FMUtoUICdrivetrain

Den nye versjonen er praktisk talt identisk med den gamle. Den eneste funksjonelle endringen er to valgfrie diagnostiske modulasjons-innganger i `step_fm`, brukt til å eksitere tårnmodene direkte:

- `_te_mod_factor` ganger koblingsmomentet (`GenSpdOrTrq`) med en faktor som driveren setter hvert steg, og tvinger dermed drivverksmomentet ved en valgt frekvens.
- `_epc_mod_factor` ganger den etterspurte elektriske effekten (`ElecPwrCom`). Med ROSCO (`VSContrl=5`) ignoreres det eksterne momentet, mens `ElecPwrCom` er kommandoen kontrolleren faktisk reagerer på, så dette representerer en effektsvingning som ROSCO oversetter til en momentvariasjon.

Begge bruker `getattr(..., 1.0)`, slik at oppførselen er identisk med den gamle filen når driveren ikke setter faktorene, og begge hoppes over i oppstarts-echo-vinduet slik at FMU-ens interne offset-kompensasjon ikke forstyrres. Alt annet (initialisering, FMU-priming, `state_derivatives`, `P_ref`, caching og terminering) er uendret.